Bollard Segmentation and Position Estimation From Lidar Point Cloud for Autonomous Mooring

Mehak Jindal, Ajit Jha, and Linga Reddy Cenkeramaddi, *Senior Member, IEEE*

***Abstract*— This article presents a computer-aided object detection and localization method from lidar 3-D point cloud data. This topic of interest is in the framework of autonomous mooring, where the ship is tied to the rigid structure on-shore (bollard) for autonomous maritime navigation. Using shape and features priors, unlike matching the whole object template to the experimental 3-D point cloud representation of the scene, two customized algorithms: 1) 3-D feature matching (3-DFM) and 2) mixed feature-correspondence matching (MFCM) are presented. The proposed algorithms discriminate and extract the 3-D points corresponding to the noncooperative bollard's surface from the background, thus capable of classification, localization, and representing it using a unique coordinate in the 3-D world. The proposed algorithms are tested and validated by implementing upon an experimental dataset of 105 scenes where the bollard is at different positions and orientations with respect to lidar mounted on the robotic arm. Statistical and probabilistic-based approaches are taken into account to determine the performance of proposed algorithms. Model parameters' estimation implies that errors resulting from the 3-DFM algorithm follow homoscedastic bimodal Gaussian distribution with individual Gaussian components having mean 0.03 and 0.09 m, and both have an equal standard deviation of 0.01 m. Furthermore, the posterior component assignment probability is used to identify and cluster the scenes that contribute to relatively larger errors. Finally, an improved algorithm, MFCM, is proposed, whose errors follow unimodal Gaussian distribution with a mean and a standard deviation of 0.03 and 0.01 m, respectively, thus mitigating the shortcomings of the former.**

***Index Terms*— Autonomous mooring, lidar 3-D point cloud, object detection, parameter estimation.**

I. Introduction

AUTONOMUS systems, such as (partial) self-driving cars, robots capable of routing in the industrial environment, drones, and unmanned vehicles capable of continuous navigation, require real-time object detection and its position estimation. However, in the aforementioned use cases, additional challenges may arise. First, the target may be uncooperative, meaning that there may not be easily recognizable markers or signs (such as reflector) on its surface. Second, it may be physically damaged and suffer optical degradation due to long exposure to the space environment. Hence, algorithms to detect the noncooperative target are challenging, and its position needs to be investigated.

This article lies in the framework of autonomous maritime navigation, in particular, focusing on autonomous mooring where the ships can govern themselves to navigate in the marine environment and Moore itself onshore without human intervention. It addresses the problem of detecting the Moore and localizing its position in 3-D such that a robotic arm can place the mooring rope around the Moore for autonomous and safe mooring (see Fig. 1). This requires dealing with three different tasks: 1) detecting the Moore; 2) determining its unique position parameter, such as (x, y, z) in the 3-D coordinate system; and 3) tracking. While the former two are used to detect the Moore and find its position in 3-D when no prior knowledge of the scene is available, the latter is used to update the position parameters on the basis of previous measurements as soon as the new measurement is available or in case the relative position of the Moore and the robotic arm is changed due to unseen conditions, such as tide and wave.

Normally, ships or vessels use GPS to locate the location of the bollard, and mooring is done manually with human involvement. The use of GPS for bollard localization has serious implications. First, they have to rely on the third part. Second, the measurement or localization from GPS is not accurate compared to local measurement using active sensors onboard in a ship or a vessel. Third, the update rate of the GPS is lower compared to lidar (proposed here). Fourth, the received signal from GPS might undergo several reflections before reaching the receiver onboard, causing multipath errors and delay. Furthermore, the GPS receiver on the ship, for example, uses multiple sets of satellites in order to find the position. Changes in satellite configuration, orientation, or position will affect the measurement. Based on the faulty position, if we guide the robot to that location with uncertainties, there is every chance that the manipulator or end-effector on the robot does not successfully Moore or even collide with the bollard or nearby structure (in the worst case) [1].

To overcome these drawbacks, it is desirable to have local sensors that are capable of determining the bollard's position effectively. For that, we propose a lidar-based sensor that lies onboard connected to the robot for mooring operations. The onboard sensor system for localization and segmentation had added advantages.

1) *Independence:* The driver of the ship or vessel does not have to rely on the third party.

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Mehak Jindal and Linga Reddy Cenkeramaddi are with the Autonomous and Cyber-Physical System (ACPS) Group, Department of Information and Communication Technology, University of Agder, 4879 Grimstad, Norway (e-mail: mehakjindal1998@gmail.com; linga.cenkeramaddi@uia.no).

Ajit Jha is with the Department of Engineering Sciences, University of Agder, 4879 Grimstad, Norway (e-mail: ajit.jha@uia.no).

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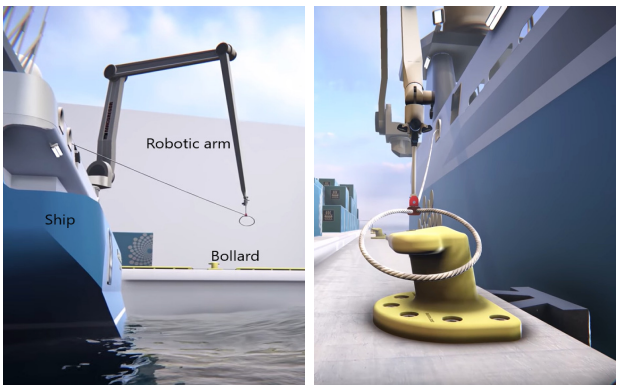


Fig. 1. Autonomous mooring. (Left) Conceptual presentation showing the relative position of ship, robotic arm, and bollard. (Right) Exploded view (with permission from MacGregor Norway AS).

2) *Redundancy:* Since the proposed methodology is based on only lidar, redundant sensors can be kept as a backup in case one fails.
3) *Local Effect:* Unlike the GPS-based prior, where the signal can be delayed or poor estimation (because of factors mentioned above), the lidar-based system is only affected by the local environment unlike the thousands of kilometers between GPS satellite and GPS receiver.
4) *Better Accuracy in Measurement:* Since the lidar is close to the bollard, the disturbances in measurement is local unlike the thousands of kilometer between GPS satellite and GPS receiver.

These problems are dealt with by designing innovative algorithms capable of processing 3-D point cloud data from active lidar. To overcome this limitation, lidars are used to sense the environment exploiting the fact they work relatively well independently of the illumination conditions and the ability to easily discriminate the target from the background. This research focuses on the use of lidar for autonomous mooring. Hence, we propose algorithms capable of detecting and localizing the bollard by means of an experiment consisting of a robotic arm equipped with scanning lidar and its relative motion with the bollard carried out in a purposely developed realistic laboratory environment. Critical aspects include the robustness explained in terms of mean error and standard deviation of the error of the proposed algorithm to detect and localize bollards at different relative orientations with respect to lidar mounted on the robotic arm. The novelties proposed here are listed in the following.

1) Propose a custom algorithm for noncooperative object (bollard) detection and localization—3-D feature matching (3-DFM) algorithm. It extracts the features of the bollard's surface (unlike the whole object) from the computer-aided model and matches those features to discriminate the experimental 3-D lidar volumetric raw point clouds' data corresponding to the bollard's surface from the background in the scene.
2) Propose the mixed feature-corresponding matching (MFCM) algorithm. It is based on the principle to uses 3-D lidar volumetric raw point clouds' data corresponding to the bollard's surface extracted in 1 using the features to identify to bollard's surface in the scene. Since it mixes two methods, feature matching as in (1) augmented by the correspondence matching in the rest of the experimental acquired scene, hence the name MFCM.
3) Both the algorithms are implemented upon 105 scenes where bollard is at different orientation relative to lidar. Detailed performance evaluation of both algorithms 3-DFM and MFCM qualitatively and quantitatively is based on probabilistic parameter estimation.

Hence, we present an end-to-end pipeline: 1) proposing algorithms to detect and segment bollards from scene; 2) validating them upon 105 experiment data; and 3) determining the performance of each of the proposed algorithms.

This article is organized as follows. Section II presents the current state of the art for noncooperative object detection, localization, and tracking, focusing on autonomous maritime navigation. Section III describes the experimental setup to mimic the realistic scenario where the lidar mounted on the robotic arm is used to capture and perceive the scene, including a bollard with various relative orientations between the lidar and the bollard. Section IV explains the customized algorithms for noncooperative bollard detection and localization in the sparse 3-D point cloud representation of the scenes. Furthermore, Section V presents the results followed by the performance evaluation of the proposed algorithms by analyzing the critical aspects, such as errors and reliability. Finally, discussion and conclusion in Section VI conclude this article.

II. Related Work

The lidar points are basically a range scan in 3-D (x, y, z). It contains the spatial coordinates of the object in 3-D making it easier to obtain the shape and pose of the object. In particular, in the case of an autonomous vehicle where the perception and control both lie on the same platform, accurately localizing the object in the 3-D coordinate system is crucial for subsequent path planning and control. To estimate the depth information in addition to the dimension of the object, a 3-D bounding box in a point cloud in lidar, such as PointNet [2], has been implemented; however, it requires extensive computing resources to train. Exploiting the sparsity of 3-D lidar point cloud, their density and resolution algorithms based on a neural network, such as Frustum PointNet [3] multiview 3-D (MV3-D) networks [4], RoarNet [5], and AVOD [6], have been implemented. Shajahan *et al.* [7] used a multiview convolution neural network with self-attention to classify roof from the lidar point cloud. However, they are computationally expensive and require loads of data for training. Furthermore, the model is highly data-dependent, which prevents its widespread application.

With the objective to overcome the limitations posed by individual methods—detect and localize the bollard in 3-D space avoiding the use of extensive processing and without a need to train the 3-D convolution model with loads of test data—numerous approaches are proposed for object detection and classification. Bayram *et al.* [8] proposed filtering of lidar point clouds using the spectral graph-based approach to

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separate the objects from the ground. Ghamisi and Höfle [9] proposed classification of the objects from raw 3-D lidar data by extracting several features, such as aggregated local point neighborhoods, laser echo ratio, variance of point elevation, plane fitting residuals, and echo intensity in addition to lidar digital surface model (DSM). Further, Damkjer and Foroosh [10] presented methods to generate lidar points to meet a certain level of details while keeping the balance between the computation cost to process 3-D lidar data and the storage space required to store sparse 3-D lidar data while optimizing the coverage volumes. Similarly, Yu *et al.* [11] proposed an algorithm based on 3-D shape context by defining the 3-D point cloud object. This method included several intermediate steps, such as road segmentation, ground segmentation, clustering similar points based on Euclidean distance, and determining Histogram. They contain several modules and the performance of the algorithm depended upon the efficiency of each of the intermediate steps. Xu *et al.* [23] proposed to detect the road curb based on a two-step method: first, determine the potential points corresponding to curb based on the energy function, and then, refine them using the least-cost path model.

In addition to the abovementioned techniques to process the sensory data and extract the object, model-based algorithms have considered serious attraction [12]–[14]. It is based on the assumption that the lidar measurement data contain the features and related information related to the objects to be detected. For example, the lidar-based 3-D point cloud data contain the range of the objects in the 3-D space, and hence, by processing them, the 3-D surface of the object can be extracted. Thus, the template matching problem can be broken down to identifying the computer-aided designed (CAD) generated model (of the object, in this case, bollard) with that in the experimental lidar acquired data points. Approaches based on point tuple matching require that the relative orientation point tuple has been proposed by Drost *et al.* [15]. The voting-based approach [16] requires additional intensity data. Hinterstoisser *et al.* [17] presented template matching based on color and depth from Kinect sensors. They propose using color information in order to prune invalid candidate transformations, while depth data are used to improve pose estimation using iterative closest point (ICP). Li *et al.* [18] devised a method for detection and placement of CAD models in scanned scenes. This method has a drawback as it is used to match the entire object in the scene, and the computation cost increases with the number of points in the scene and/or increasing the number of templates in the database. Liu *et al.* [19] used template matching together with kernel density estimation clustering to detect the pedestrian. Here, the raw 3-D point cloud needed to be preprocessed, such as ground segmentation, grid filtering, hierarchical segmentation, and projection of 3-D data points to 2-D plane using the principal component analysis (PCA). This projection of information to lower dimension results in loss of information. In addition, it is not always easy to separate the ground and hierarchical segmentation in itself is computationally expensive as it requires comparison of each data point with neighboring points.

There are numerous works done related to mooring operations [20]–[22]. However, they are focused on manual operation operations and how the structural integrity can be maintained, the effect of mooring angle on mooring operations, and so forth. In regards to the autonomous mooring where the Moore is detected by sensors and mooring operation is carried autonomously, very limited work has been done. Perkovic *et al.* [24] proposed laser-based berthing and mooring of the ship without any towing assistance. Furthermore, Leite *et al.* [25] presented a sensor fusion-based approach for autonomous docking operation of surface vehicles (SVs). However, the approach was based on fusing data from lidar, IMU, and GPS to determine the geometric features of the floating platform and then determine the relative position and orientation of Moore relative to SV.

Continuing the similar trend, we present novel methods to detect and localize the bollard in the scene from the raw 3-D point cloud. Overcoming the abovementioned drawbacks, our algorithm (3DFM) first extracts the features in the model data and then uses it to filter the data points in the experimental scene. This, on the one hand, lowers the computational cost, as the feature extraction in the model has to be done only once, while lowering the number of data points for comparisons in the experimental data. Furthermore, an improved algorithm (MFCM) is presented by extending the 3-DFM algorithm to extract the point clouds and features of the bollard from the experimental scene and use it as a template to detect and localize the bollard in the rest of the scenes.

III. Problem Formulation

The concept behind the autonomous mooring is shown in Fig. 1. It consists of a robot (with manipulator) equipped with sensors (in this case, lidar) and a perception algorithm to perceive the scene. It does this in discrete steps of time in which it moves some predefined vertical distance from the bollard. The robot repeats this process until it localizes and finds the unique coordinate of the bollard in the 3-D space. Once the position of the bollard is found, the robotic arm holding the Moore navigates to the bollard and lays off the Moore around the bollard to dock the ship. The conceptual diagram in Fig. 1 is replicated in the laboratory (as shown in Fig. 2). The experimental setup consists of Ouster lidar fitted on an ABB robot placed at a vertical distance of 1.4 m from the bollard. The bollard is manufactured in-house[1] in the way that it meets the requirement of uncooperative target: 1) it is built with foam material; 2) it is not tagged with any kind of markers; 3) it is not equipped with any kind of reflecting material; and 4) it is not painted with bright colors instead painted in yellow to have a low-moderate reflection. Then, the robotic arm is moved at different orientations relative to the bollard, and the point cloud is acquired at each point (represented by a blue dot in Fig. 3). A total of scenes,

[1]The CAD model and the measurement of dimension of template bollard are available at https://github.com/ajitjha14/Bollard-Segmentation-and-Position-Estimation-fromLidar-Point-Cloud-for-autonomous-mooring/upload/main/Bollard_model_dimension

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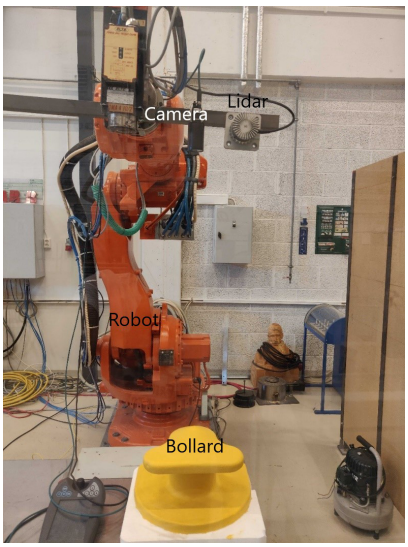


Fig. 2. Experimental setup for detection and classification of bollard for autonomous mooring. In this setup, the vertical distance of the lidar relative to bollard is 1.4 m. The lidar frame is given as follows: $+x$ is right, $+y$ is up, and $+z$ is out of plane.

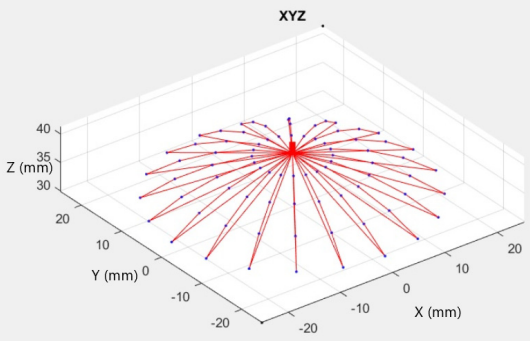


Fig. 3. Data acquisition at different relative positions of robot with respect to the bollard.

$N_s = 105$, were acquired[2] During each measurement, the lidar is configured to have vertical and horizontal resolutions of 64 and 1024, respectively, thus giving the measurement points of $64 \times 1024 = 65\,536$ for each scene. Then, each of these acquired scenes is tested upon algorithms to detect the bollard (represented by N_b points) and represent it by a unique coordinate in 3-D, $\boldsymbol{p} \in \mathbb{R}^3$.

The notations are as given follows. Vectors and point clouds are defined by small, italics bold face, constant quantities by capital normal font, variables by small italics normal font, and set of objects enclosed in curly braces. Let $\mathcal{S} = \{\boldsymbol{s}_i\}_{i=1}^{N_s}$, where $N_s \in \mathbb{Z}$ be the set of all scenes captured during the experiment. Each element of $\mathcal{S}$ (i.e., $\boldsymbol{s}_i$) is coordinate of 3-D sparse point containing information about the scene. Mathematically, $\boldsymbol{s}_i = \{\boldsymbol{p}_j\}_{j=1}^{N_p}$, where $\boldsymbol{p}_j \in \mathbb{R}^3 \ \forall i$. The objective is to detect the bollard $b_i = \{\boldsymbol{p}_k\}_{k=1}^{N_b}$, where $\boldsymbol{p}_k \in \mathbb{R}^3, N_b < N_p \ \forall i$ from the scene s_i, finally presenting it with a unique coordinate $o_i = \{\boldsymbol{p}_i\}$, where $\boldsymbol{p}_i \in \mathbb{R}^3 \ \forall i$. For the ease of understanding the detailed relationships among the scene dataset ($\mathcal{S}$), individual scene (e.g., $\boldsymbol{s}_i$) and the coordinates of each lidar point ($\boldsymbol{p}_j$) are described in Fig. 4.

[2]The dataset is available at git: https://github.com/ajitjha14/Bollard-Segmentation-and-Position-Estimation-fromLidar-Point-Cloud-for-autonomous-mooring/tree/main/Data. At the time of uploading, Git only allows 100 files to upload, so 100 out of 105 files can be found here.

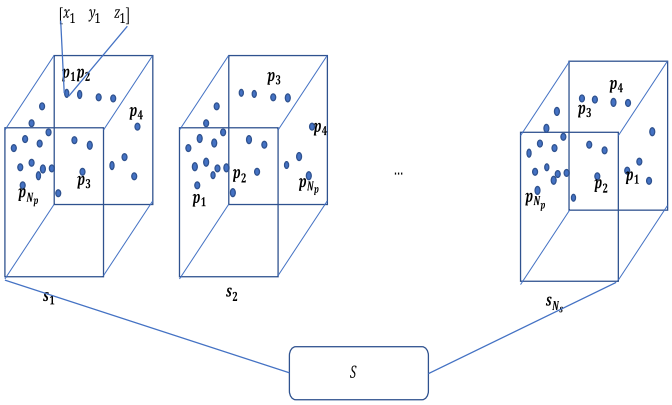


Fig. 4. Relationship between scene set, ($\mathcal{S}$), scene ($\boldsymbol{s}_i$), and the coordinates of points in the scene ($\boldsymbol{p}_i$). The dimension of $\mathcal{S}$ is $[1 \times N_s]$, $\boldsymbol{s}_i$ is $[1 \times N_p]$, and $\boldsymbol{p}_i$) is $[3 \times 1]$. For the experiment, $N_S = 105$, and $N_p = 65\,536$.

IV. Algorithms

To detect the bollard in the scene and determine its unique coordinate in 3-D from the sparse point cloud captured by lidar, we employ two methods: 1) 3-DFM and 2) MFCM.

1) 3-DFM: In the case of 3-DFM, first, the CAD model of the bollard (T) is converted to point cloud ($\boldsymbol{t_c}$). As explained above, we have a custom-built 3-D CAD model of bollard that was manufactured in-house. While manufacturing the CAD model, the parameters were chosen in a way that it fits with the real-world bollard (especially in Norway). Since the actual dimension of the bollard, e.g., its length, width, and height, is known, using these geometric parameters, a model of plane is formulated using the RANSAC algorithm [26]. Then, from this model, the points lying on the plane are differentiated from others. In this way, the points corresponding to the top surface of the bollard ($\boldsymbol{t}$) are discriminated against from the rest of the point cloud. Once the point corresponding to the plane of the top surface of the bollard is estimated, it is matched with the experimental dataset ($\mathcal{S}$, as defined above in Figs. 2 and 3) obtained from lidar for registration and to identify the bollard under consideration. Furthermore, the plane of the top surface of the bollard ($\boldsymbol{t_p}$) and its centroid ($\boldsymbol{o}$) are extracted representing the bollard by a unique coordinate. The entire procedure for template matching-based extraction from the scene is described in Algorithm 1.

2) MFCM: In the case of MFCM, first, the bollard and the lidar are calibrated such that the latter lies vertically at a distance of 1.4[3] m, for example, above the former. Under these conditions, the scene is acquired (s_1, as shown in Fig. 2 and described by the center point in Fig. 3). Using Algorithm 1, the point cloud corresponding to the top surface of the bollard in the experimental scene ($\boldsymbol{s}_1$) is obtained and is used as a template ($\boldsymbol{t_t}$) to locate the bollard and find the centroid of the top surface of the bollard to represent it by a unique coordinate in rest of the scenes $\{\boldsymbol{s}_i\}_{i=2}^{N_s-1}$. All the steps associated with

[3]The choice of vertical distance sets the distance where the lidar starts scanning.

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template extraction from the calibrated scene and its use in correspondence matching are explained in Algorithms 2 and 3, respectively.

Algorithm 1 Bollard Detection and Position Estimation Using 3-DFM

Input: Scenes, $\mathcal{S} = \{\boldsymbol{s}_i\}_{i=1}^{N_s}$; Template, T
Output: Surface of bollard, $\boldsymbol{t}_{\boldsymbol{p}'i}$; centroid of surface of bollard, $\boldsymbol{o}_i$; error, e_i $\quad \forall i$
1: $\boldsymbol{t}_{\boldsymbol{c}} \leftarrow$ convert T to point cloud
2: $\boldsymbol{t} \leftarrow$ point cloud representing the surface of bollard obtained from geometric dimension and features of surface of bollard from $\boldsymbol{t}_{\boldsymbol{c}}$
3: $c \leftarrow$ centroid of the surface of bollard obtained from $\boldsymbol{t}$
4: **for** i=1 to N_s **do**
5: $\quad \boldsymbol{s}_i \leftarrow$ remove all the points in $\boldsymbol{s}_i$ with +ve y coordinate ▷ +y represents the point above the lidar
6: $\quad \boldsymbol{s}_i \leftarrow$ denoise $\boldsymbol{s}_i$
7: $\quad \boldsymbol{t}_{\boldsymbol{p}i} \leftarrow$ plane of the bollard in $\boldsymbol{s}_i$ using the features in $\boldsymbol{t}$ and distance 1.4 m ▷ the robot starts looking for bollard as the distance is 1.4 m from lidar (for e.g.)
8: $\quad \theta_i \leftarrow$ angle between the $\boldsymbol{t}_{\boldsymbol{p}i}$ and $\boldsymbol{t}$ using Iterative closest point algorithm (ICP)
9: $\quad \boldsymbol{t}_{\boldsymbol{p}i} \leftarrow$ rotated $\boldsymbol{t}_{\boldsymbol{p}i}$ by angle θ_i to align extracted plane with the template, $\boldsymbol{t}$
10: $\quad \boldsymbol{o}_i \leftarrow$ centroid of the surface of bollard point clouds obtained from $\boldsymbol{t}_{\boldsymbol{p}i}$
11: $\quad e_i \leftarrow$ error, distance between c and $\boldsymbol{o}_i$ calculated using Euclidean norm
12: $\quad$ **return** $\boldsymbol{t}_{\boldsymbol{p}i}$, $\boldsymbol{o}_i$, e_i
13: **end for**

Algorithm 2 Template for MFCM

Input: Calibrating scene, $\boldsymbol{s}_1$; surface of the bollard, $\boldsymbol{t}$ (from Algorithm 1, step 2)
Output: Surface of bollard, $\boldsymbol{t}_{\boldsymbol{t}}$ ▷ returns the surface of bollard extracted from calibrating scene ($\boldsymbol{s}_1$)
1: $\boldsymbol{s}_1 \leftarrow$ remove all the denoised points in $\boldsymbol{s}_1$ with +ve y coordinate ▷ +y represents the point above the lidar
2: $\boldsymbol{t}_{\boldsymbol{t}} \leftarrow$ plane of the bollard in $\boldsymbol{s}_1$ using $\boldsymbol{t}$ ▷ using RANSAC and ICP algorithm
3: **return** $\boldsymbol{t}_{\boldsymbol{t}}$

V. Results

The detailed investigation on the efficiency of the methodologies present is further explained.

A. 3-DFM

Fig. 5 summarizes the results for the 3-DFM method (see Algorithm 1), starting from the acquired scene [see Fig. 5(a)], 3-D CAD template of bollard [see Fig. 5(b)], and the point cloud corresponding to the top surface of the bollard [see Fig. 5(c)], and finally, the points clouds corresponding to the bollard's top surface are extracted using Algorithm 1

Algorithm 3 Bollard Detection and Position Estimation Using MFCM

Input: Scenes, $\mathcal{S} = \{\boldsymbol{s}_i\}_{i=2}^{N_s}$; Template, $\boldsymbol{t}_{\boldsymbol{t}}$ (obtained from Algorithm 2)
Output: Surface of bollard, $\boldsymbol{t}_{\boldsymbol{p}}$; centroid of surface of bollard, $\boldsymbol{o}_i$; error, e_i $\quad \forall i$
1: **for** i=1 to N_s **do**
2: $\quad$ repeat step 5 to 12 (described in Algorithm 1)
3: $\quad$ **return** $\boldsymbol{t}_{\boldsymbol{p}i}$, $\boldsymbol{o}_i$, e_i
4: **end for**

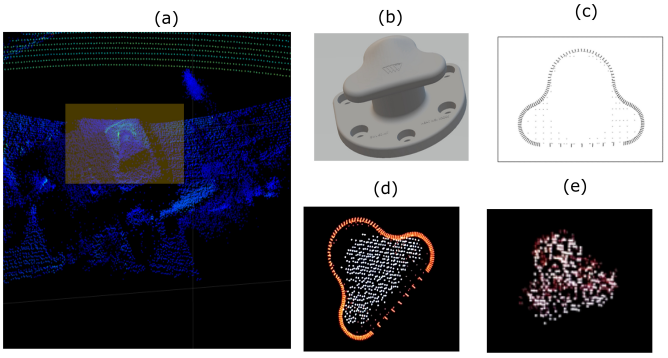


Fig. 5. Testing and validation of the proposed algorithms. (a) Experimental scene acquired from the setup defined in Fig. 2 and the bollard highlighted by shaded rectangle. (b) 3-D template of bollard. (c) Top surface of bollard extracted. (d) Extraction of the bollard's surface from the scene using the 3-DFM using Algorithm 1 (white) and compared against the template (orange) (e) using MFCM, Algorithm 2 and 3 (white) and compared against the template (red).

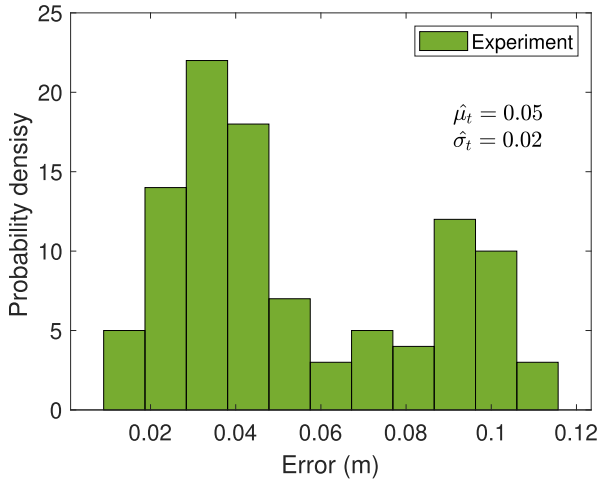


Fig. 6. Performance evaluation of 3-DFM (see Algorithm 2). The errors incurred by processing the set of scenes in $\mathcal{S}$ follow univariate bimodal Gaussian distribution.

[see Fig. 5(d)]. Algorithm 1 is applied upon all scenes $\mathcal{S}$ with different orientations, as shown in Fig. 3, using setup described in Fig. 2. The errors defined as the Euclidean norm (see Algorithm 1, step 11) are summarized in Fig. 6. It is basically the histogram of the errors defined in terms of probability density. It is observed that the probability density function (pdf) of the error incurred from the 3-DFM algorithm is bimodal as it has two dominant modes. Thus, this case can be classified as a univariate bimodal Gaussian mixture

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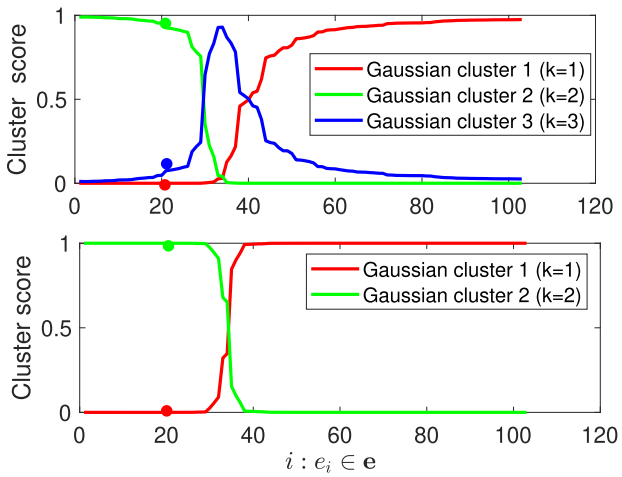


Fig. 7. Optimal choice of Gaussian components.

model (GMM). Hence, the problem now translates to: 1) finding the optimal Gaussian components (cluster), K, which make this distribution; 2) estimation of parameters of individual Gaussian components $\hat{\theta}^t = [\hat{\mu}^t \ \hat{\sigma}^t]$; and 3) probability that a given value from the error dataset falls within given Gaussian cluster.

Mathematically, a univarite multimodal GMM is defined in terms of individual Gaussian component weights (ϕ_k), mean (μ_k), and variances (σ_k).[4] For example, a GMM with K Gaussian components, the kth Gaussian component, has mean μ_k and variance σ_k and has weight ϕ_k. Since we have the data related to errors, it is desirable to calculate the posterior probability, defined as the probability that the error e_i ($e_i \in \boldsymbol{e}$) belongs to a particular Gaussian component k with probability p. Mathematically, it is defined as

$$p(e_i) = \sum_{k=1}^{K} \phi_k \mathcal{N}(x \mid \mu_k, \sigma_k) \tag{1}$$

where

$$\mathcal{N}(e_i \mid \mu_k, \sigma_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(e_i-\mu_k)^2}{2\sigma_k^2}} \tag{2}$$

and

$$\sum_{k=1}^{K} \phi_k = 1. \tag{3}$$

1) Gaussian Component: Though, two dominant modes in the probability density of the error (see Fig. 6) are clearly visible; the quantitative analysis is performed to ascertain the optimal number of Gaussian cluster (modes). For this, mathematical induction-based approach is taken. First, the error data are fitted with three Gaussian components, $k = 3$, using the $k-$means++ algorithm [27]. The cluster score, defined as the probability that a given error sample from $\{e_i\}_{i=2}^{N_s}$ lies in kth cluster, is plotted against the scene ($i \in [2, \ldots, N_s]$)[5] Fig. 7 (top) shows the results for $k = 3$. It is observed that, for most of the cases, the Gaussian component with $k = 2$ (C_2) is almost zero for most of the scenes, and only the other Gaussian components, $k = 1$ (C_1) and $k = 3$ (C_2), are dominant. For example, consider the case that, when $i = 20$ (demonstrated by the solid filled circle), the cluster scores of components $C_1(k = 1)$, $C_2(k = 2)$, and $C_3(k = 3)$ are 0, 0.1, and 0.9, respectively. Similarly, Fig. 7 (bottom) shows the results for $k = 2$. In this case, the Gaussian clusters are well defined with minimum overlap. For example, for scene $i = 20$, the (demonstrated by the solid filled circle) the cluster scores of components $C_1(k = 1)$ and $C_2(k = 2)$ are 0 and 1, respectively. Thus, it is binary clustering for most of the cases (except at the transitions) with one cluster having dominant score over other.

[4]since the model here is univariate, we stick with the notation corresponding to univariate, bimodal GMM.

[5]It should be noted that each scene number, i, corresponds to different orientation of lidar with respect to bollard, as described in Fig. 3. Indexing of scene, i, starts from 2 as the first index ($i = 1$) is used as template for rest of the scenes.

Algorithm 4 Parameter Estimation of the GMM Resulting From Errors in 3-DFM (see Algorithm 1)

Input: $\mathbf{e} = \{e_i\}_{i=2}^{N_s}$ ▷ error data
$K = 2$ ▷ number of Gaussian components (please see section V(a) titled *Gaussian component*)
Output: $\hat{\boldsymbol{\mu}}^t, \hat{\boldsymbol{\sigma}}^t, \hat{\boldsymbol{\phi}}^t$, ▷ mean, standard deviation and mixture weight of individual Gaussian component respectively (see (1), (2) and (3)) such that $\hat{\boldsymbol{\mu}}^t, \hat{\boldsymbol{\sigma}}^t, \hat{\boldsymbol{\phi}}^t \in \mathbb{R}^K$

$\mathbf{r} \leftarrow K$ random integers between 1 and N ▷ $\mathbf{r} \in \mathbb{Z}^K$

$\hat{\boldsymbol{\mu}}^t \leftarrow \mathbf{e}(\mathbf{r})$ ▷ assign random values from error data to the estimated mean of each Gaussian component

$\hat{\boldsymbol{\sigma}}^t \leftarrow \frac{1}{N}\sum_{i=1}^{N}(e_i - \bar{\mathbf{e}})^2$ ▷ assign sample variance to the estimated variance of each Gaussian component

$\hat{\boldsymbol{\phi}}^t \leftarrow \frac{1}{K}$ ▷ Set all Gaussian components distribution prior estimates to the uniform distribution

for $iter = 1$ to 500 **do**

$\hat{\boldsymbol{\gamma}}^t \leftarrow \frac{\hat{\boldsymbol{\phi}}^t \mathcal{N}(\mathbf{e} \mid \hat{\boldsymbol{\mu}}^t, \hat{\boldsymbol{\sigma}}^t)}{\sum_{\forall i}(\hat{\boldsymbol{\phi}}^t \mathcal{N}(\mathbf{e} \mid \hat{\boldsymbol{\mu}}^t, \hat{\boldsymbol{\sigma}}^t)}$ ▷ $\hat{\boldsymbol{\gamma}}^t \in \mathbb{R}^{N\times K}$. $\hat{\gamma}_{ik}^t$ is the probability that i^{th} value of the error data, e_i is generated by Gaussian component C_k. Thus $\hat{\gamma}_{ik}^t = p(C_k \mid e_i, \hat{\boldsymbol{\mu}}^t, \hat{\boldsymbol{\sigma}}^t, \hat{\boldsymbol{\phi}}^t)$

$\hat{\boldsymbol{\phi}}^t \leftarrow \frac{\sum_{\forall k} \hat{\boldsymbol{\gamma}}^t}{N}$ ▷ update Gaussian component weight

$\hat{\boldsymbol{\mu}}^t \leftarrow \frac{\sum_{\forall k} \hat{\boldsymbol{\gamma}}^t \mathbf{e}}{\sum_{\forall k} \hat{\boldsymbol{\gamma}}^t}$ ▷ update mean

$\hat{\boldsymbol{\sigma}}^2 \leftarrow \frac{\sum_{\forall k} \hat{\boldsymbol{\gamma}}^t (\mathbf{e} - \hat{\boldsymbol{\mu}}^t)^2}{\sum_{\forall k} \hat{\boldsymbol{\gamma}}^t}$ ▷ update variance

return $\hat{\boldsymbol{\mu}}^t, \hat{\boldsymbol{\sigma}}^t, \hat{\boldsymbol{\phi}}^t$

end for

2) Parameter Estimation of Individual Gaussian Component: The algorithm to estimate the parameters and the probability is explained in Algorithm 4. The results are described in Figs. 8 and 9. It is observed that the numerical method-based approach to determine the parameters works effectively for all the parameters. We summarize the parameters by $\hat{\boldsymbol{\theta}}^t = [\hat{\boldsymbol{\mu}}^t; \hat{\boldsymbol{\sigma}}^t; \hat{\boldsymbol{\phi}}^t]$, such that $\hat{\boldsymbol{\theta}}^t \in \mathbb{R}^{3\times 2}$, as the univariate bimodal GMM is defined by three parameters, and each of parameters has two values as determined above

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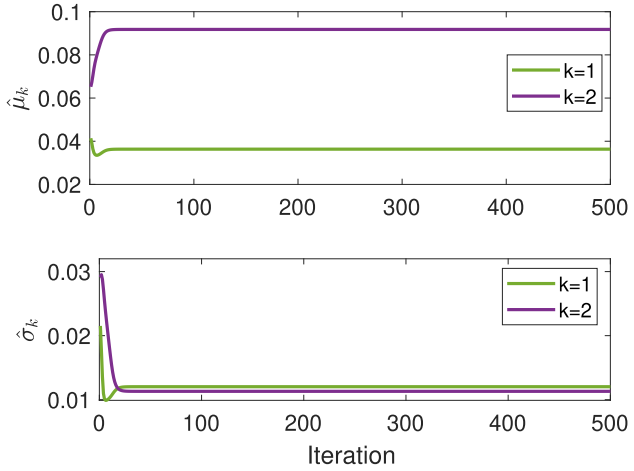


Fig. 8. Error parameter estimation resulting from 3-DFM described in Algorithm 4. The errors follow univariate, bimodal GMM. (Top) Mean and (Bottom) standard deviation of individual Gaussian components are - $\hat{\boldsymbol{\mu}}^t = [\hat{\mu}_1, \hat{\mu}_2] = [0.03, 0.09]$ m and $\hat{\boldsymbol{\sigma}}^t = [\hat{\sigma}_1, \hat{\sigma}_2] = [0.01, 0.01]$ m, respectively.

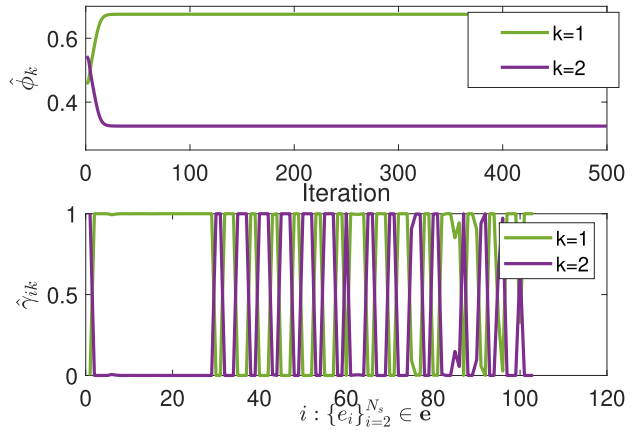


Fig. 9. Error parameter estimation resulting from 3-DFM described in Algorithm 4. The errors follow univariate, bimodal GMM. (Top) Weight of component C_1 and C_2 is 0.67 and 0.33, respectively. (Bottom) Probability that error sample e_i comes form kth component with parameters $\hat{\boldsymbol{\mu}}^t$, $\hat{\boldsymbol{\sigma}}^t$, and $\hat{\boldsymbol{\phi}}^t$ explained above.

in Algorithm 4. To be very specific, the parameters are given as follows: $\hat{\theta}_1^t = [\hat{\mu}_1^t \ \hat{\sigma}_1^t \ \hat{\phi}_1^t] = [0.03 \ 0.01 \ 0.67]$ and $\hat{\theta}_2^t = [\hat{\mu}_2^t \ \hat{\sigma}_2^t \ \hat{\phi}_2^t] = [0.09 \ 0.01 \ 0.33]$. Furthermore, the observed experimental error approximated by a Gaussian components is shown in Fig. 10 for probabilistic description and performance estimation. It can be clearly observed that the parameters estimated from Algorithm 4 fit well with the experimental observed error from template matching. Hence, the error distribution resulting from the template matching algorithm follows univariate bimodal GMM.

3) Clustering Gaussian Components: Having determined the model parameters and using Bayes' theorem, the probability that the data points belong to cluster k is determined as

$$p(C_k \mid e_i) = \frac{\hat{\phi}_k^t \mathcal{N}(e_i \mid \hat{\mu}_k^t, \hat{\sigma}_k^t)}{\sum_{j=1}^{j=K} \hat{\phi}_j^t \mathcal{N}(e_i \mid \hat{\mu}_j^t, \hat{\sigma}_j^t)} \tag{4}$$

where $k \in [1, \ldots, K]$. All the related parameters are obtained and described above (please see Sections V-A and V-B). Putting the values of those parameters in 4, the probability that individual error values (e_i) lie to a particular Gaussian

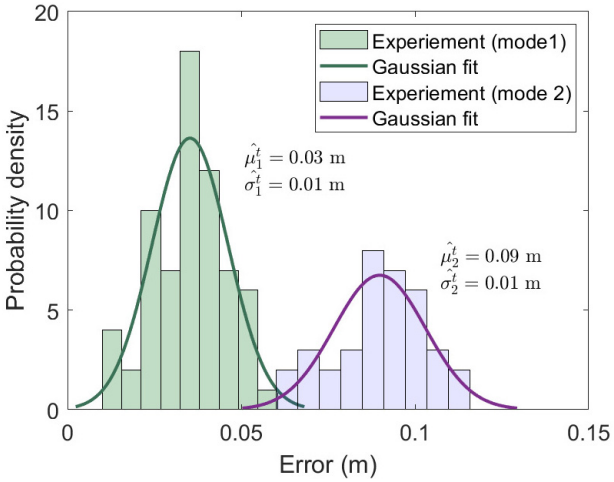


Fig. 10. Performance evaluation of the 3-DFM algorithm. From the probability density (histogram), it is observed that the distribution has two modes; hence, the errors follow the univariate bimodal GMM. Parameters of Gaussian components are obtained from Algorithm 4 with the parameters summarized in Figs. 8 and 9 compared with the experimental error. It is observed that the experimental error well fits univariate bimodal GMM.

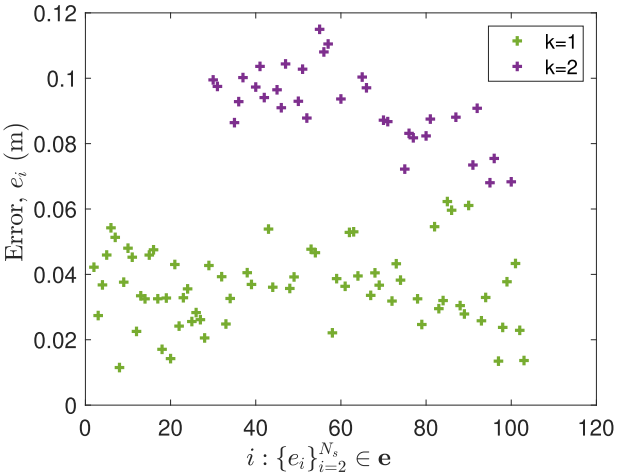


Fig. 11. Clustering the error samples resulting from template matching algorithm described in Algorithm 4. Putting the parameters obtained from Figs. 8 and 9 in (4), we are able to separate the samples that corresponds to Gaussian component having lower error values ($k = 1, C_1$) from the relatively higher error values ($k = 2, C_2$).

component (cluster) is determined and shown in Fig. 11. The scenes i and, hence, the orientation of lidar relative to the bollard that causes a larger error (purple) are well separated by their counterpart that causes relatively lower errors (green).

B. MFCM

Fig. 5(e) summarizes the results for the MFCM method (see Algorithm 1 and 2). Similar to 3-DFM, the histogram in terms of probability density as a function of error is shown in Fig. 12 (blue). It is observed that the pdf of the error incurred from the MFCM algorithm is unimodal as it has one dominant mode. Thus, this case can be classified as univariate unimodal Gaussian model (unlike the 3-DFM case, where it was GMM). Furthermore, it is desirable to estimate the parameters of statistical model that best describes the probability of occurrence of observed error such that statistical

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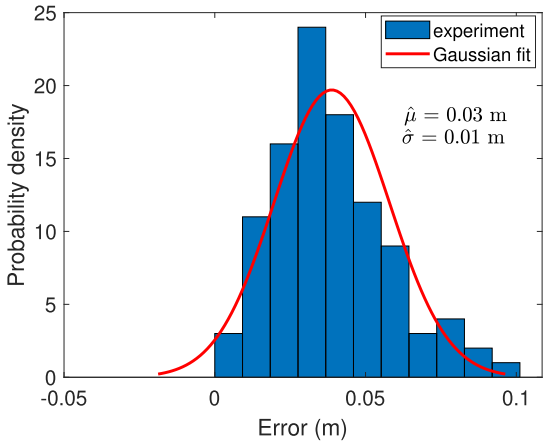


Fig. 12. Errors resulting from the correspondence matching algorithm (3). The error follows a Gaussian pdf with mean ($\mu^c = 0.03$ m) and standard deviation ($\sigma^c = 0.01$ m).

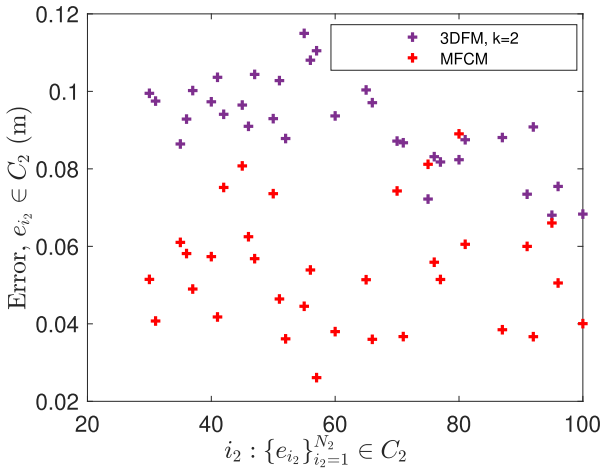


Fig. 13. Performance comparison of 3-DFM (see Algorithm 1) with the MFCM algorithm (see Algorithms 2 and 3). The errors introduced by the former are compensated by the latter. This result can be attributed to the absence of extra mode with mean $\hat{\mu}_2^t$ in latter.

inference can be made. Mathematically, this translates the problem to finding an optimal parameter θ that maximizes the statistical model. Mathematically,

$$\hat{\mu^c} = \arg\max_{\mu} \mathcal{N}(\boldsymbol{e} \mid \mu, \sigma) \tag{5}$$

$$\hat{\sigma^c} = \arg\max_{\sigma} \mathcal{N}(\boldsymbol{e} \mid \mu, \sigma). \tag{6}$$

It is found that the Gaussian model has the following parameters: estimated mean, $\hat{\mu^c} = 0.03$ m, and estimated deviation, $\hat{\sigma^c} = 0.01$ m. It is well described in Fig. 12 (red). Furthermore, t-test with significance level ($\alpha = 0.05$) is carried to perform the statistical hypothesis testing to validate the estimated parameter. Here, we define the null hypothesis (H_0) and alternate hypothesis (H_1) as -H_0 : error comes from Gaussian distribution having estimated mean, $\hat{\mu} = 0.03$ m, and H_1 : error comes from Gaussian distribution having different estimated means, $\hat{\mu} \neq 0.03$ m. From the test, the p-value (p), the probability of observing significant deviations from estimated mean ($\hat{\mu} = 0.03$) assuming that null hypothesis is true approximately, equals 1. Since $p > \alpha$, we fail to reject the null hypothesis, and thus, the errors follow Gaussian distribution with mean $\hat{\mu^c} = 0.03$ m.

Detailed modeling of the error statistics of both algorithms—3-DFM and MFCM—confirms that, while the errors incurred from former follow univariate bimodal Gaussian distribution with two components (GMM) having means of 0.03 and 0.09 m and standard deviation of 0.01 m, the latter follows univariate unimodal Gaussian distribution with a mean of 0.03 and a standard deviation of 0.01 m. Thus, the MFCM algorithm performs better in terms of lowered mean error, while both the algorithms are equally precise (standard deviation). To make a better illustration of this conclusion, referring to Fig. 11, the scene i corresponding to the component C_2, ($k = 2$) resulting from 3-DFM is compared with MFCM algorithm (as shown in Fig. 13). It is observed that the MFCM algorithm has smaller errors compared to the 3-DFM, hence giving the improved performance. It is also evident that about 95% of the error lies in the range [0.01 m, 0.05 m] ($\hat{\mu^c} \pm 2\hat{\sigma^c} = 0.03 \pm 0.02$). Since the diameter of Moore rope is much larger (approximately 2x) than the diameter of bollard (usually in terms of meters), the maximum error estimated from the correspondence matching is within the allowable error.

VI. Conclusion and Discussion

Two algorithms—3-DFM and MFCM—are presented to detect and localize a noncooperative target (bollard) by processing raw 3-D point cloud data from lidar aimed at autonomous mooring. The performance of the proposed algorithm is tested and validated upon 105 scenes where the relative position of lidar and bollard is different. Based on detailed statistical and probabilistic theories, the model parameters of the errors resulting from both algorithms are estimated. Furthermore, using Bayes' theorem and the estimated parameters, the errors are clustered in a way to find out the suitability of one algorithm over the other. Finally, it was concluded that, though both the methods had an equal mean deviation error of 0.01 m, the MFCM algorithm having a mean error of 0.03 m outperforms the 3-DFM algorithm.

Furthermore, it would also be desirable to study the effect of different lighting conditions (e.g., limited vision) and different weather conditions replicating urban sea. Second, the proposed algorithm can be extended to the bollards of different shapes, sizes, structures, and morphologies. Another aspect is to evaluate the performance of algorithms at different distances at various lidar resolutions. The slight increase in the vertical distance of lidar relative to bollard would not affect the result much. However, if the distance is increased significantly, the bollard would contain relatively a smaller fraction of total points, and the detection could be a challenge. However, this can be compensated by choosing the lidar operating mode with greater resolution, which, in turn, would increase the cost. Furthermore, a balance between lidar operating resolution (64×1024 or other configuration depending upon lidar model and cost) and its distance from bollard should be optimized, and it remains the work for the future.

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REFERENCES

[1] S. Sukkarieh, E. M. Nebot, and H. F. Durrant-Whyte, "A high integrity IMU/GPS navigation loop for autonomous land vehicle applications," *IEEE Trans. Robot. Autom.*, vol. 15, no. 3, pp. 572–578, Jun. 1999.

[2] R. Q. Charles, H. Su, M. Kaichun, and L. J. Guibas, "PointNet: Deep learning on point sets for 3D classification and segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 652–660.

[3] C. R. Qi, W. Liu, C. Wu, H. Su, and L. J. Guibas, "Frustum pointnets for 3D object detection from RGB-D data," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 918–927.

[4] X. Chen, H. Ma, J. Wan, B. Li, and T. Xia, "Multi-view 3D object detection network for autonomous driving," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1907–1915.

[5] K. Shin, Y. P. Kwon, and M. Tomizuka, "RoarNet: A robust 3D object detection based on region approximation refinement," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, 2019, pp. 2510–2515, doi: 10.1109/IVS.2019.8813895.

[6] J. Ku, M. Mozifian, J. Lee, A. Harakeh, and S. L. Waslander, "Joint 3D proposal generation and object detection from view aggregation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2018, pp. 1–8.

[7] D. A. Shajahan, V. Nayel, and R. Muthuganapathy, "Roof classification from 3-D LiDAR point clouds using multiview CNN with self-attention," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1465–1469, Aug. 2020.

[8] E. Bayram, P. Frossard, E. Vural, and A. Alatan, "Analysis of airborne LiDAR point clouds with spectral graph filtering," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 8, pp. 1284–1288, Aug. 2018.

[9] P. Ghamisi and B. Hofle, "LiDAR data classification using extinction profiles and a composite kernel support vector machine," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 659–663, May 2017.

[10] K. L. Damkjer and H. Foroosh, "Lattice-constrained stratified sampling for point cloud levels of detail," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5627–5641, Aug. 2020, doi: 10.1109/TGRS.2020.2967880.

[11] Y. Yu, J. Li, H. Guan, C. Wang, and J. Yu, "Semiautomated extraction of street light poles from mobile LiDAR point-clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1374–1386, Mar. 2015.

[12] Y. Yu, J. Li, H. Guan, F. Jia, and C. Wang, "Three-dimensional object matching in mobile laser scanning point clouds," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 3, pp. 492–496, Mar. 2015.

[13] K. Williams, M. Olsen, G. Roe, and C. Glennie, "Synthesis of transportation applications of mobile LIDAR," *Remote Sens.*, vol. 5, no. 9, pp. 4652–4692, Sep. 2013.

[14] Y. Yu, J. Li, J. Yu, H. Guan, and C. Wang, "Pairwise three-dimensional shape context for partial object matching and retrieval on mobile laser scanning data," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 5, pp. 1019–1023, May 2014.

[15] B. Drost, M. Ulrich, N. Navab, and S. Ilic, "Model globally, match locally: Efficient and robust 3D object recognition," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2010, pp. 998–1005.

[16] B. Drost and S. Ilic, "3D object detection and localization using multimodal point pair features," in *Proc. 2nd Int. Conf. 3D Imag., Modeling, Process., Vis. Transmiss. (3DIMPVT)*, Oct. 2012, pp. 9–16.

[17] S. Hinterstoisser *et al.*, "Model based training, detection and pose estimation of texture-less 3D objects in heavily cluttered scenes," in *Proc. Asian Conf. Comput. Vis.*, Daejeon, South Korea, 2012, pp. 548–562.

[18] Y. Li, A. Dai, L. Guibas, and M. Nießner, "Database-assisted object retrieval for real-time 3D reconstruction," in *Proc. Comput. Graph. Forum*, vol. 34, 2015, pp. 435–446.

[19] K. Liu, W. Wang, and J. Wang, "Pedestrian detection with lidar point clouds based on single template matching," *Electronics*, vol. 8, no. 7, p. 780, Jul. 2019.

[20] S.-W. Ji, M.-S. Choi, and Y.-B. Kim, "A study on position mooring system design for the vessel moored by mooring lines," *IEEE/ASME Trans. Mechatronics*, vol. 20, no. 6, pp. 2824–2831, Dec. 2015.

[21] S. Han, B.-Z. Jia, and Y.-M. Gu, "Study on optimizing influence of mooring angle in multi-point mooring positioning system," in *Proc. Chin. Control Decis. Conf. (CCDC)*, Jun. 2018, pp. 3777–3782.

[22] M. Dissanayake *et al.*, "Automated application of full matrix capture to assess the structural integrity of mooring chains," *IEEE Access*, vol. 6, pp. 75560–75571, 2018.

[23] S. Xu, R. Wang, and H. Zheng, "Road curb extraction from mobile LiDAR point clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 996–1009, Feb. 2017.

[24] M. Perkovič *et al.*, "Laser-based aid systems for berthing and docking," *J. Mar. Sci. Eng.*, vol. 8, no. 5, p. 346, May 2020.

[25] P. Leite, R. Silva, A. Matos, and A. M. Pinto, "An hierarchical architecture for docking autonomous surface vehicles," in *Proc. IEEE Int. Conf. Auto. Robot Syst. Competitions (ICARSC)*, Apr. 2019, pp. 1–6.

[26] M. Fischler and R. Bolles, "Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography," *Commun. ACM*, vol. 24, no. 6, pp. 381–395, 1981.

[27] D. Arthur and S. Vassilvitskii, "k-means++: The advantages of careful seeding," in *Proc. 18th Annu. ACM-SIAM Symp. Discrete Algorithms*. Philadelphia, PA, USA: SIAM, 2007, pp. 1027–1035.



Mehak Jindal received the bachelor's degree in computer science and engineering from BML Munjal University, Kapriwas, India, in 2020. She is pursuing the master's degree in remote sensing and geographic information systems with the Indian Institute of Remote Sensing, Dehradun, India.

She was a Visiting Student at the Department of ICT, University of Agder, Agder, Norway, from January 2020 to June 2020. Her primary interest areas include the application of machine learning for satellite images and remote sensing.



Ajit Jha received the B.Sc. degree in electronics and communication engineering from Khulna University, Khulna, Bangladesh, in 2007, the European master's degree in photonic networks from Aston University, Birmingham, U.K., and the Scuola Superiore Sant Anna, Pisa, Italy, in 2012, and the Ph.D. degree from the Technical University of Catalunya, Barcelona, Spain, and the Karlsruhe Institute of Technology, Karlsruhe, Germany, in 2016.

From 2016 to 2019, he worked at various industries related to autonomous vehicle working on innovative technologies, such as automotive Ethernet, ADAS, surround view systems, camera mirror systems, and blind sport warning to name a few. He is an Associate Professor of mechatronics at the Department of Engineering Sciences, University of Agder, Grimstad, Norway. He is also member of Top Research Center Mechatronics (TRCM) and the ACPS Group. He has (co)authored more than 20 articles and two patents. He is actively involved in research focused on sensors, sensor fusion, image/signal processing, ML, ADAS functionalities toward autonomous systems, and IoT.

Dr. Jha was a recipient of Erasmus Mundus Masters Course (EMMC) and Erasmus Mundus Joint Doctorate (EMJD) both funded by European Union (EU). He has been an active reviewer, and a member of technical program committee of numerous international peer-review journals and conferences.



Linga Reddy Cenkeramaddi (Senior Member, IEEE) received the master's degree in electrical engineering from the Indian Institute of Technology, New Delhi, India, in 2004, and the Ph.D. degree in electrical engineering from the Norwegian University of Science and Technology (NTNU), Trondheim, Norway, in 2011.

He worked for Texas Instruments, Dallas, TX, USA, in mixed signal circuit design before joining the Ph.D. Program at NTNU. After finishing his Ph.D., he worked in radiation imaging for an atmosphere space interaction monitor (ASIM Mission to International Space Station) at the University of Bergen, Bergen, Norway from 2010 to 2012. At present, he is the Group Leader of the autonomous and cyber-physical systems (ACPS) and working as an Associate Professor with theUniversity of Agder, Campus Grimstad, Norway. His main scientific interests are in cyber-physical systems, autonomous systems, and wireless embedded systems.

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